Andrew Olvera

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RESEARCH INTERESTS

Agile, high-speed robotics in complex and GNSS-denied environments. Integrating robot learning, optimal control, computer vision, and learned world models for robust perception, planning, and coordination across multi-platform robot teams.

EDUCATION

York College of Pennsylvania Aug 2023 – May 2026
B.S. in Computer Science, York, PA

- Coursework: Data Structures, Algorithms, Linear Algebra, Calculus, Probability & Statistics, Computer Vision, Computer Graphics

Cabrini University Aug 2021 – May 2023
Undergraduate coursework, Radnor, PA; transferred to York College

RESEARCH EXPERIENCE

Starscream: Supervised Pretraining for Route-Conditioned Agile Flight 2026 – Present
Independent research

- Pretrained a 5.84M-parameter causal transformer policy using MPCC expert labels, DAgger, DART, and an auxiliary task-delta prediction head.
- Conditioned the policy on recent state and control history, six ordered gates, privileged vehicle settings, and a test-time speed command.
- In a three-seed eval100 comparison, achieved 66.7% clean-and-timely success with the transformer, versus 52.5% for LSTM and 50.1% for history MLP; inputs and training budgets differed.
- Implemented RL fine-tuning of pretrained flight policies; evaluating benefits of pretraining against matched baselines.

Tells: Opponent Beliefs from Play History in a MicroRTS World Action Model 2026 – Present
Independent research

- Built action-conditioned latent world models and state/action tokenizers for MicroRTS, with complete-state and fog-visible representations.
- Distilled a 16-code opponent belief from fog-of-war history using privileged training targets; improved return prediction, while downstream policy benefits remained inconclusive.
- Trained PMPO actor-critic policies with eight-step imagined rollouts in a frozen learned world, alongside real-engine data collection; compared belief-conditioned and separately trained no-belief policies.

Reward Design in Small-Model RLVR 2025 – 2026
Independent research

- Studied RLVR on Qwen3 0.6B (GSM8K), comparing GRPO and DAPO across group sizes, then binary, structured, and rank-based rewards on a fixed baseline.
- Exploratory single-seed runs reached roughly 69–72% accuracy from a 68% SFT baseline; tracked reward variation as binary rewards saturated.
- Analyzed token-level advantages, dynamic clipping, and reward structure to study when the optimizer retains a useful learning signal.

MANUSCRIPTS IN PREPARATION

A. Olvera. *Starscream: Supervised Pretraining for Route-Conditioned Agile Flight.* In preparation, 2026.

A. Olvera. *Tells: Opponent Beliefs from Play History in a MicroRTS World Action Model.* In preparation, 2026.

A. Olvera. *Reward Design in Small-Model RLVR.* In preparation, 2026.

HONORS

- Dean's List, York College of Pennsylvania · Fall 2025, Spring 2026

INDUSTRY EXPERIENCE

Sapio Sciences May 2025 – Present

Application Engineer Intern, York, PA

- Develop and deliver customer-specific software solutions, combining hands-on engineering with customer-facing implementation.
- Collaborate across teams to translate customer needs into working solutions and coordinate delivery.

Dataforma May 2025 – Jan 2026

Software Engineer Intern, Web Development, York, PA

- Developed full-stack web applications with Vue.js and Kotlin, including refactoring legacy JSP modules into a separate frontend and backend.
- Worked across development and QA teams to implement features, debug defects, execute test plans, and validate deployments.

TECHNICAL SKILLS

ML & RL: PyTorch, Hugging Face Transformers, TRL, vLLM; imitation learning (DAgger, DART), policy gradients (PPO, GRPO, DAPO), RLVR, RLHF/DPO, model-based RL, self-supervised learning, computer vision

Control & Simulation: Control theory, MPC/MPCC (acados), ROS, Flightmare, Isaac Sim, CARLA, Basilisk, MATLAB/Simulink

Languages: Python, C++, C, Java, Kotlin, JavaScript, C#, SQL

Tools: Linux, Git, Docker, Bash, HDF5, WebDataset